

演算法實驗作業四

Lab of Algorithm Assignment 4

凸包演算法比較分析：Monotone Chain 與分而治之法

Comparative Analysis of Convex Hull Algorithms: Monotone Chain and Divide-and-Conquer

課程名稱 / Course	演算法實驗 / Lab of Algorithm
作業類型 / Type	演算法分析報告，不需撰寫完整程式 Algorithm analysis report; full programming is not required
核心主題 / Theme	Monotone Chain 與 Divide-and-Conquer Convex Hull 的虛擬碼、流程、複雜度與比較分析 Pseudocode, workflow, complexity, and comparative analysis of Monotone Chain and Divide-and-Conquer Convex Hull

一、前導說明

1. Introduction

在演算法設計中，分而治之 (Divide and Conquer) 是一種核心方法：先將大問題拆成較小的子問題，分別求解後，再把子結果合併成原問題的答案。這種方法不只出現在排序與搜尋，也廣泛應用於計算幾何。

In algorithm design, Divide and Conquer is a core paradigm: a large problem is split into smaller subproblems, each subproblem is solved independently, and the partial results are then merged into a final solution. This idea appears not only in sorting and searching, but also broadly in computational geometry.

本次作業以平面點集的凸包 (Convex Hull) 問題為主題。給定一組平面上的點，凸包是能夠包住所有點的最小凸多邊形。若把點想像成木板上的釘子，再把橡皮筋套在最外圍，最後收縮形成的邊界，就是這組點的凸包。

This assignment focuses on the convex hull problem for planar point sets. Given a set of points in the plane, the convex hull is the smallest convex polygon that encloses all of them. If the points are imagined as nails on a board and a rubber band is stretched around the outside, the boundary formed after the band tightens is the convex hull.

凸包是計算幾何中的經典問題，常見於影像處理、路徑規劃、碰撞偵測、地理資訊系統與形狀分析等應用。更重要的是，不同凸包演算法展現了不同的演算法設計觀點：有些方法倚重排序與掃描，有些方法則清楚體現分而治之的結構。

The convex hull is a classical problem in computational geometry and appears in image processing, path planning, collision detection, GIS, and shape analysis. More importantly, different convex hull algorithms

reveal different design perspectives: some rely on sorting and scanning, while others clearly embody the divide-and-conquer paradigm.

本次作業不要求同學實作完整程式，而是要求從虛擬碼、流程解釋、圖解示意、時間複雜度與方法比較的角度完成分析。重點不在語法或除錯，而在於你是否真正理解演算法的步驟、邏輯與差異。

This assignment does not require students to implement a full program. Instead, the work should be completed through pseudocode, process explanation, diagrams, time-complexity analysis, and method comparison. The emphasis is not on syntax or debugging, but on whether you genuinely understand the steps, logic, and differences of the algorithms.

二、作業目的

2. Objectives

- 理解凸包問題的幾何意義與實際應用。

Understand the geometric meaning and practical applications of the convex hull problem.

- 熟悉 Monotone Chain 的流程與關鍵判斷。

Become familiar with the workflow and key decisions in Monotone Chain.

- 理解 Divide-and-Conquer Convex Hull 的分割、遞迴與合併概念。

Understand the splitting, recursion, and merging concepts in Divide-and-Conquer Convex Hull.

- 學會從虛擬碼層面分析演算法，而不只依賴程式實作。

Learn to analyze algorithms at the pseudocode level rather than relying only on implementation.

- 比較兩種方法在設計思想、步驟、複雜度與理解難度上的差異。

Compare the two methods in terms of design philosophy, steps, complexity, and conceptual difficulty.

三、作業題目

3. Assignment Task

請比較以下兩種凸包演算法：

(1) Monotone Chain

(2) Divide-and-Conquer Convex Hull

你需要以演算法分析報告的方式完成作業，說明兩者的核心思想、流程、虛擬碼、複雜度，以及它們在方法設計上的主要差異。

Compare the following two convex hull algorithms:

(1) Monotone Chain

(2) Divide-and-Conquer Convex Hull

You are required to complete this assignment as an algorithm analysis report, explaining the core ideas, workflows, pseudocode, complexity, and major design differences between the two methods.

四、作業要求

4. Assignment Requirements

- 本次作業不需要撰寫完整、可執行的程式。

This assignment does not require a complete executable program.

- 兩種演算法都必須提供清楚的虛擬碼。

Clear pseudocode must be provided for both algorithms.

- 每段虛擬碼後都必須搭配你自己的文字說明。

Each pseudocode section must be accompanied by your own written explanation.

- 報告中必須包含圖解或示意圖，幫助說明演算法流程。

The report must include diagrams or illustrations to explain the algorithmic process.

- 除了分別介紹兩種方法之外，還必須進行完整比較分析。

In addition to describing each method separately, you must provide a full comparative analysis.

五、報告內容要求

5. Report Content Requirements

章節 Section	必含內容 (中文)	Required Content (English)
1	問題定義：說明什麼是凸包、為何稱為最小凸多邊形，以及其應用或意義。	Problem definition: explain what a convex hull is, why it is the smallest convex polygon, and its applications or significance.
2	Monotone Chain：說明為何先排序、如何建立 lower hull 與 upper hull、外積或轉向判斷的角色，並附虛擬碼。	Monotone Chain: explain why sorting comes first, how the lower and upper hulls are constructed, the role of cross product or turn testing, and include pseudocode.
3	Divide and Conquer：說明如何切分點集、如何遞迴求解、如何合併左右局部凸包，以及合併步驟為何關鍵，並附虛擬碼。	Divide and Conquer: explain how the point set is split, how recursion works, how the left and right local hulls are merged, why merging is critical, and include pseudocode.
4	比較分析：比較兩者的設計思想、流程、是否符合分而治之、理解難度、實作難度、優點與限制。	Comparative analysis: compare the design philosophy, workflow, divide-and-conquer characteristics, conceptual difficulty, implementation difficulty, strengths, and limitations.
5	時間複雜度：分析兩種方法的主要時間成本，說明排序、遞迴與合併在整體成本中的角色。	Time complexity: analyze the major time costs of both methods and explain the roles of sorting, recursion, and merging in the total complexity.

6	結論與個人觀點：整理你的理解與觀察，說明你認為哪一種方法較直觀、較能展現分而治之思想。	Conclusion and personal reflection: summarize your understanding and observations, and explain which method you find more intuitive and which better demonstrates divide-and-conquer.
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六、格式要求

6. Formatting Requirements

- 繳交格式限 PDF 或 DOCX。

Accepted submission formats are PDF or DOCX only.

- 建議報告長度為 4 至 8 頁。

The recommended report length is 4 to 8 pages.

- 請使用清楚可讀的字型、明確的標題層級與整齊一致的中英文排版。

Use a clear and readable font, a clear heading hierarchy, and a neat and consistent bilingual layout.

- 圖示需有標題或說明；虛擬碼需排版清楚，不可整段擠在一起。

Each figure must have a title or caption, and pseudocode must be clearly formatted rather than crowded into one block.

- 檔名請使用：學號_姓名_Assignment4.pdf 或 StudentID_Name_Assignment4.docx。

Use the filename format: StudentID_Name_Assignment4.pdf or StudentID_Name_Assignment4.docx.

七、繳交內容

7. Submission Items

- 完整報告一份。

One complete report.

- 報告中需包含：問題定義、兩種演算法的虛擬碼與說明、圖解、複雜度分析、比較分析與結論。

The report must include: problem definition, pseudocode and explanation for both algorithms, diagrams, complexity analysis, comparative analysis, and conclusion.

八、重要限制

8. Important Restrictions

- 不得只貼上網路或 AI 生成內容而不加整理與說明。

Do not simply paste content from the Internet or AI tools without organization and explanation.

- 不得只附圖片版虛擬碼而沒有文字解釋。

Do not submit image-only pseudocode without textual explanation.

- 比較分析必須是完整內容，不可只列零散關鍵字。

Comparative analysis must be written as complete content rather than a list of scattered keywords.

- 若使用 AI 工具協助整理內容，你仍需對正確性與表達品質負責。

If you use AI tools to help organize your work, you remain responsible for correctness and quality of expression.

九、評分方式

9. Grading Criteria

項目 Item	說明 Description	比重 Weight
問題理解與演算法說明 Problem understanding and algorithm explanation	是否正確說明凸包問題，並清楚描述兩種演算法。 Whether the convex hull problem is explained correctly and both algorithms are clearly described.	25%
虛擬碼品質 Pseudocode quality	步驟是否完整、易讀，且有搭配說明。 Whether the steps are complete, readable, and properly explained.	20%
分而治之分析 Divide-and-conquer analysis	是否真正理解分割、遞迴與合併的意義。 Whether the meaning of splitting, recursion, and merging is truly understood.	20%
比較分析與複雜度 Comparison and complexity	比較是否具體，複雜度分析是否合理。 Whether the comparison is specific and the complexity analysis is reasonable.	20%
報告表達與格式 Presentation and formatting	結構完整、版面整齊、中英文內容清楚。 Whether the structure is complete, the layout is neat, and the bilingual content is clear.	15%

十、總結

10. Summary

本次作業的重點不在於把程式寫出來，而在於理解並說清楚兩種凸包演算法背後的设计思想。Monotone Chain 展現了排序與幾何掃描的思維，而 Divide and Conquer 則體現了分而治之在計算幾何中的應用。希望你能透過本次作業，看見演算法不只是程式碼，更是一種拆解問題、組織步驟與建立邏輯的方法。

The focus of this assignment is not on writing code, but on understanding and clearly explaining the design ideas behind two convex hull algorithms. Monotone Chain demonstrates sorting and geometric scanning, while Divide and Conquer embodies the divide-and-conquer paradigm in computational geometry. Through this assignment, you should see that an algorithm is not merely code, but a way of decomposing problems, organizing steps, and building logical structure.